

tTEscanR: : CHEAT SHEET

Overview

tTEscanR is a powerful, versatile and user-friendly R package designed to quantify and analyze the relationship between codon usage in mRNA and the availability of corresponding anticodons in tRNA. The package computes a **theoretical translation efficiency (tTE)** score as a proxy of translation efficiency. As a result, both gene expression and chromatin accessibility profile data are required.

The analysis can be carried out across three hierarchical layers of information: gene expression, codon and anticodon pool, and amino acid level. This multi-layered approach provides a comprehensive view of translation efficiency.

With its modular structure, the **tTEscanR** package allows to run specific components independently or as part of a comprehensive pipeline `Run_tTEscanR_pipeline()`, offering flexibility to enhance and complement the analysis of codon-anticodon dynamics across various biological contexts.

Data specifications

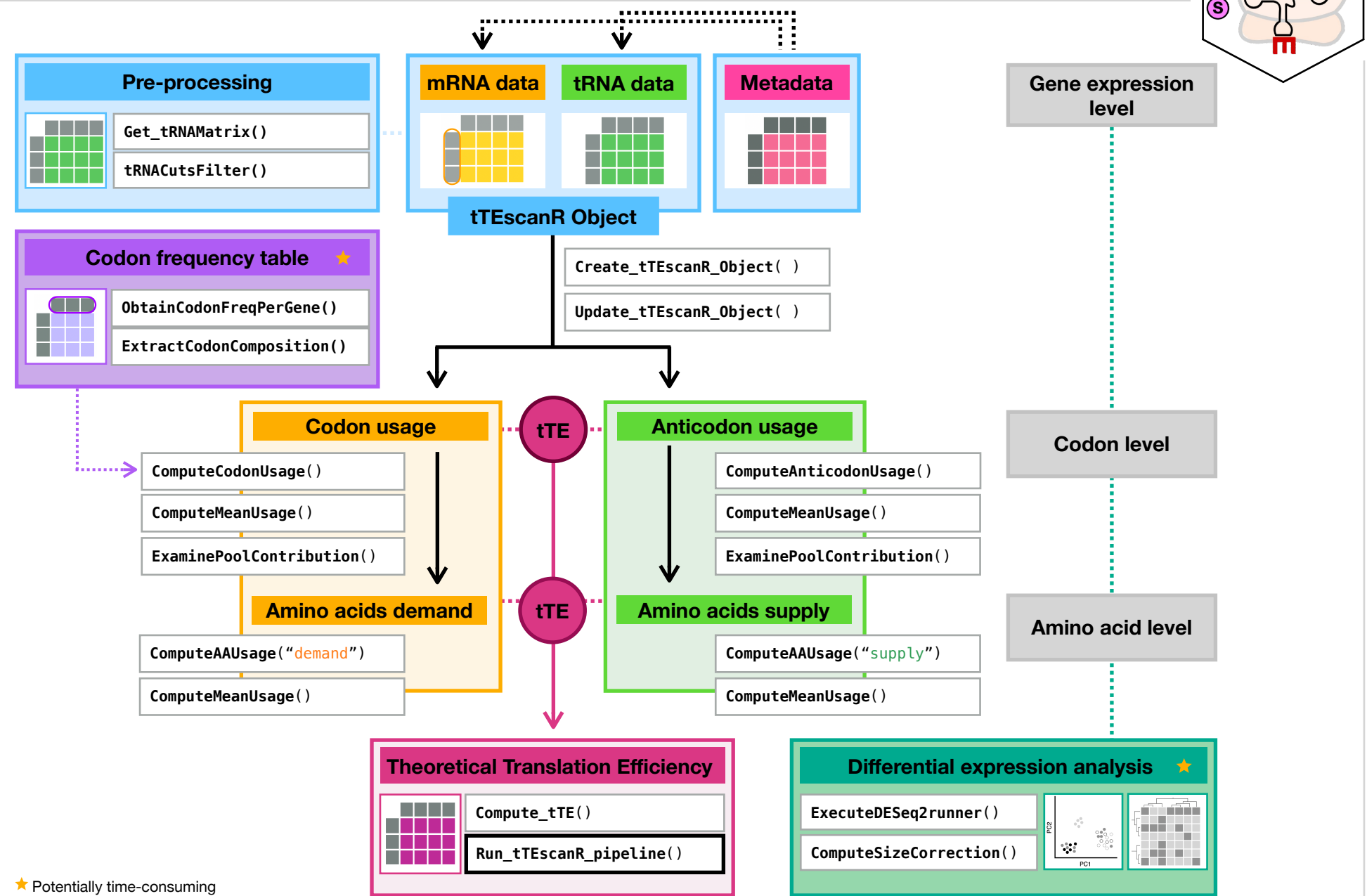
The accepted mRNA and tRNA inputs consist of pre-processed gene expression count matrices, with **features** as rows and **conditions** as columns. The package is optimized for **bulk** and **single-cell** datasets.

mRNA genes and confidently predicted tRNA genes are considered as features. In bulk datasets, conditions typically represent a combination of model and replicate, whereas in single-cell datasets, conditions correspond to specific tissue and cell type designations.

Codon frequency per gene table

The codon usage assessment in **tTEscanR** is performed through matrix multiplication between the mRNA gene expression data and the codon frequency per gene table. This table represents the codon distribution of each protein-coding gene in a reference genome. **tTEscanR** includes pre-computed tables for human (hg38) and mouse (mm39) genomes, which can be directly used in `ComputeCodonUsage()`.

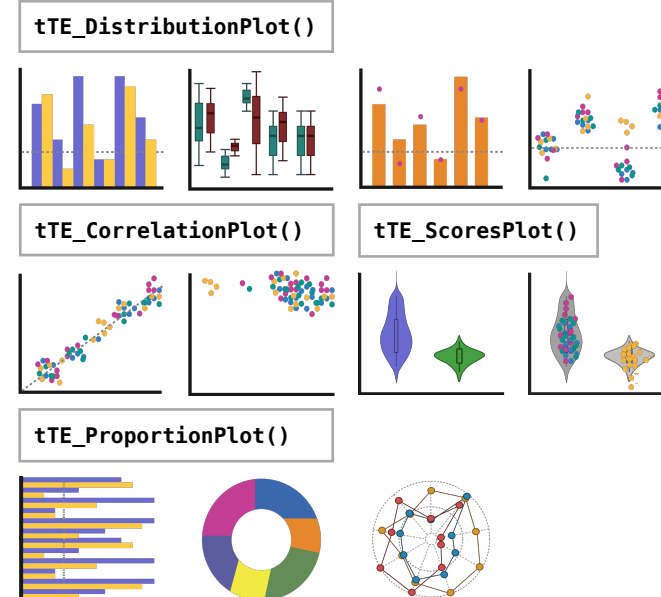
Additionally, **tTEscanR** allows users to retrieve codon frequency per gene matrices of any reference organism available in Ensembl through `ObtainCodonFreqGene()`, or to directly provide a reference genome or sequences of interest. This flexibility enables broad generalization and applicability of the computations across a wide range of species.



Visualization module

tTEscanR includes a dedicated visualization module with multiple functions for generating plots from the output tables produced at each step of the analysis.

The visualization module enhances intuitive and simplified result interpretation, making easier to explore and understand the data.



Installation instructions

```
install.packages("/avarassanchez/tTEscanR")  
library(tTEscanR)
```

Reference

Cell type-specific analysis by single-cell profiling identifies a stable mammalian tRNA-mRNA interface and increased translation efficiency in neurons.

William Gao, Carlos J Gallardo-Dodd, Claudia Kutter